

What a flood depth is worth, in damage

The estimand

Expected annual damage (EAD), as a fraction of asset value, for a portfolio of European power-generation assets: for each JRC return period, read the modelled flood depth at the site, map it to a damage fraction with a published depth-damage curve, then integrate that fraction over the exceedance-probability curve to get one annual-average number per site.

The answer

Capacity-weighted mean EAD across 167 plants in 9 countries: 0.81% of asset value per year -- interval [0.14%, 1.67%] from resampling locations, [0.13%, 1.54%] combining location resampling with curve choice drawn once per draw. 7 of 167 sites show any flood depth at all; the number is small and skewed, which is exactly why it carries an interval instead of a single figure.

Why this, not a heavier number

A risk manager does not want a depth map. They want one number per asset that feeds a loss distribution, and this module is the full chain from a public hazard raster to that number, done honestly about where the uncertainty is.

Data and method, nothing typed

Hazard

JRC/Copernicus EFAS river flood hazard maps for Europe and the Mediterranean Basin, v3.1.1 (Baugh et al. 2024): nine return-period depth rasters (10, 20, 30, 40, 50, 75, 100, 200, 500 years), 90m ~300MB each. `src/flood_depth_damage.py` never downloads a full raster -- it reads the TIFF directory with a handful of small HTTP range requests, then fetches only the compressed tiles under the portfolio's coordinates.

Depth-damage curve

Huizinga, de Moel & Szewczyk (2017), JRC105688, the published Europe curves, fetched from CLIMADA's open-source redistribution of the JRC database and parsed out of that source file's literal arrays -- the infrastructure sector as the primary curve, industrial and commercial as the published alternates.

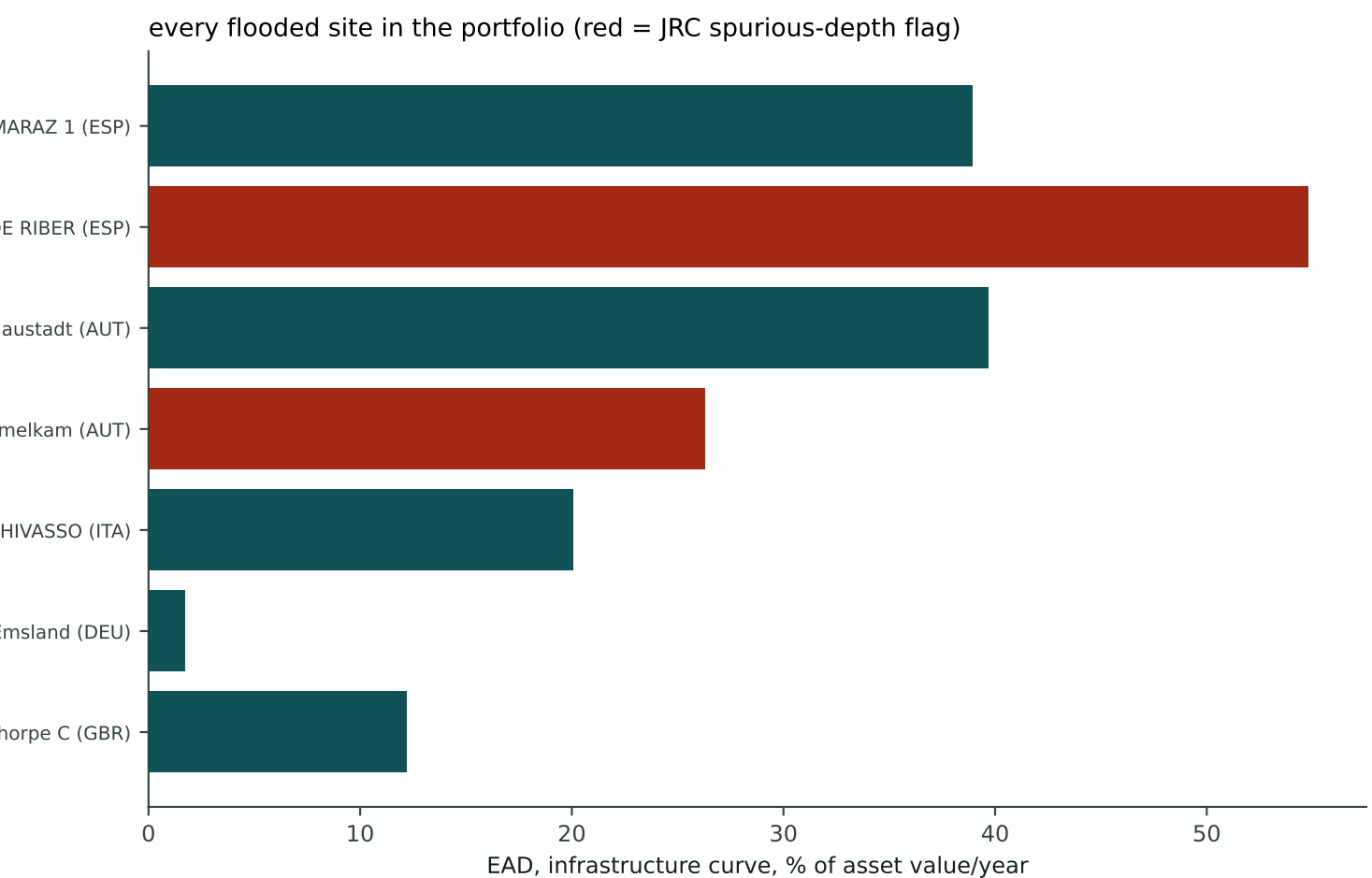
Portfolio

WRI Global Power Plant Database, the 20 largest plants by capacity in each of 9 countries with a documented history of damaging river floods, excluding hydro and wave/tidal plants: those sit in the river or sea by design, so the hazard map's modelled depth at their coordinates is the normal operating water level, not flood damage to a building -- see page 5.

Result: a handful of sites carry the whole number

What is actually flooded

7 of 167 sites show nonzero depth at any return period, all of them gas or nuclear plants sited on a river for cooling water -- never a coal, solar or wind site in this portfolio. The headline EAD is a capacity-weighted average over all 167 sites, so these 7 carry the entire result.



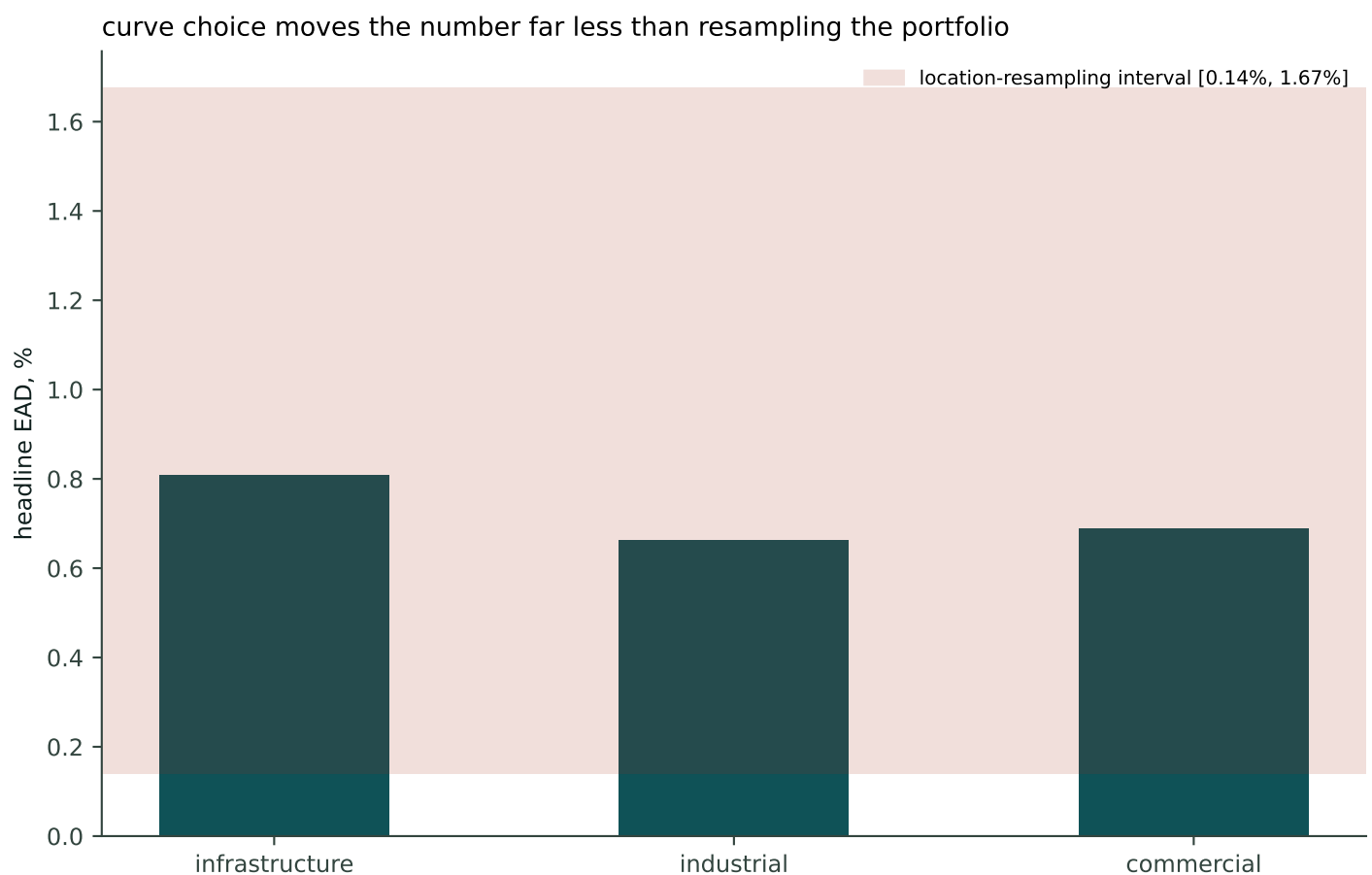
Sensitivity: the curve, or the portfolio?

The roadmap's question

Repeat with at least two published curves and report the spread; say whether the hazard or the curve drives the answer. Swapping the primary curve (infrastructure, 0.808%) for industrial (0.662%) headline by 0.146% points at most.

What actually dominates here

Resampling which plants land in the portfolio moves the number by 1.532% points -- 10.5x the curve's range. With or without the survey design, not the damage function: this portfolio's driver is hazard/location resampling, not the curve choice. The curves published for Europe's building sectors turn out to be close to each other at the depths this portfolio actually sees.



Sensitivity and validation

The EAD integral has its own arbitrary choice

Whether to assume zero damage at an annual probability of 1 (so floods more frequent than the rarest tabulated return period still contribute something) or to simply stop the integral at the most frequent return period in the data moves the headline from 0.808% to 0.178% bigger than the curve choice on the previous page. Extending the rarest point's damage flat below RP500 instead of truncating there moves the headline by only 0.0047% points -- that end of the integral is not where the sensitivity lives.

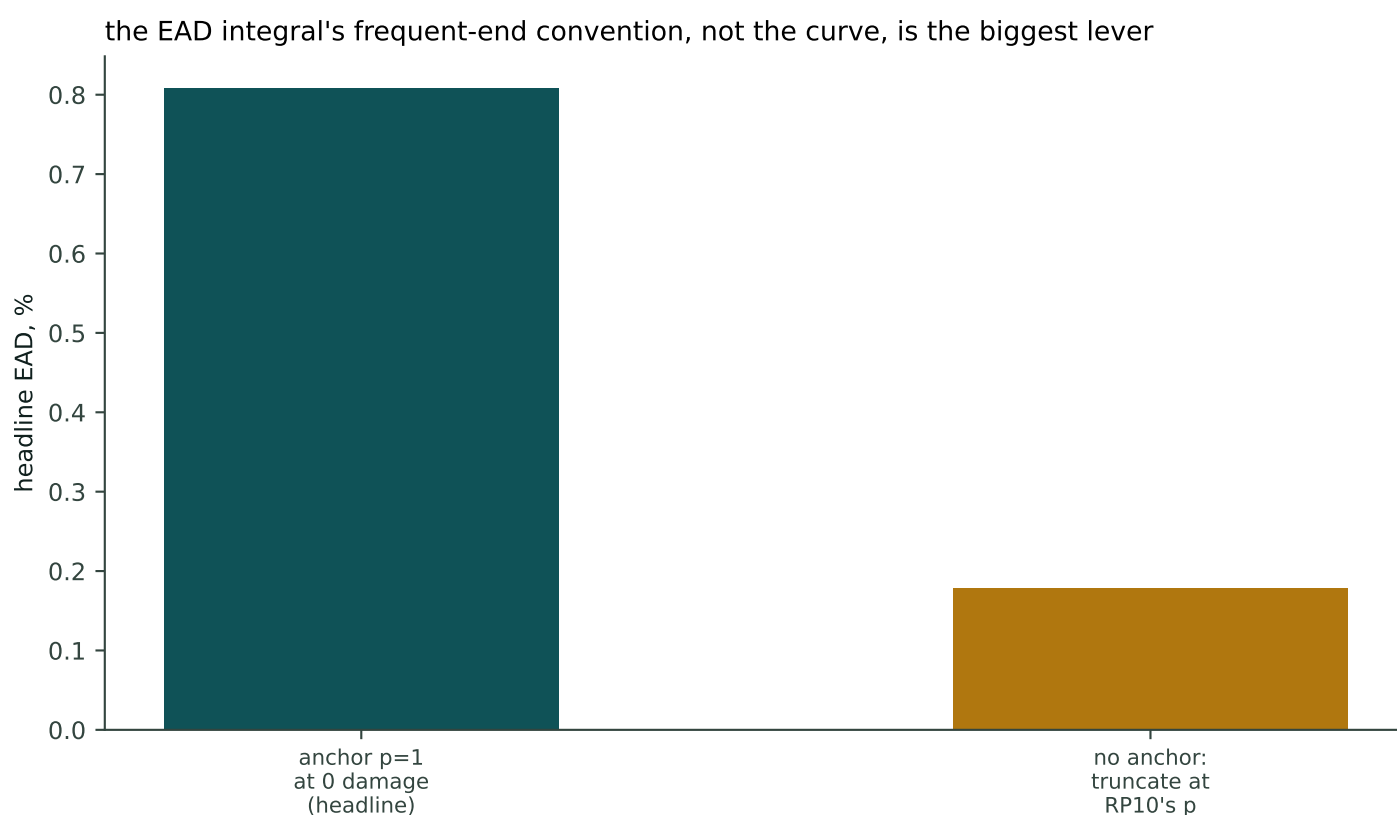
Portfolio size

Taking the 10/15/20/30 largest plants per country instead of 20 moves the headline across 0.81%, 0.62%, 0.81%, 1.1% magnitude throughout, no reversal of the finding.

Validation: the JRC's own quality flags

2 of 7 flooded

sites fall inside JRC's own 'spurious depth areas' layer -- cells where the model may overpredict depth in small channels -- flagged red on page 3 rather than hidden. Separately, 1 site(s) sat on the hazard map's permanent-water-bodies patch (a constant fill value, not a flood signal) and were screened to zero rather than counted as the most exposed site in the book. No independent EAD or loss benchmark was fetched for this module; none of comparable public, code-fetchable granularity was found -- stated here rather than skipped.



Limits, and what I would build next

The assumption that breaks the conclusion

A depth-damage curve is a national average for a generic building in its sector. It says nothing about this specific plant's finished-floor elevation, flood defences, or whether critical equipment sits in a basement or on a raised platform -- two sites at the same modelled depth can have very different real damage. This module answers 'what does a hazard map say' not 'what would this specific asset actually lose', and the gap between the two is largest for exactly the handful of flooded sites that carry the headline.

No backtest, and why not one for this module

STANDARD.md requires backtesting anything forward-looking. The JRC hazard maps used here are a present-day statistical hazard assessment (LISFLOOD/LISFLOOD-FP fitted to the historical EFAS reanalysis), not a climate-change projection -- there is no future year to hold out and no projected trend fitted by this module, so there is nothing to backtest. A forward-looking version would need JRC's climate-adjusted hazard maps, which this release does not include.

Asset coordinates are a point, not a footprint

WRI's plant coordinates are a single point per site; a large plant's actual equipment can span several 90m pixels with different depths. The portfolio also excludes hydro by construction (page 2) -- this module says nothing about flood risk to hydroelectric generation, only to thermal and nuclear plants.

Next

Replace the single-point coordinate with the plant's actual footprint where one is public, and bring in the JRC flood-risk (not just hazard) layer if a country-level EAD benchmark becomes available to validate the headline number against, which page 5 could not do this round.